HOMEWORK DAY:03

29 August 2023 10:02

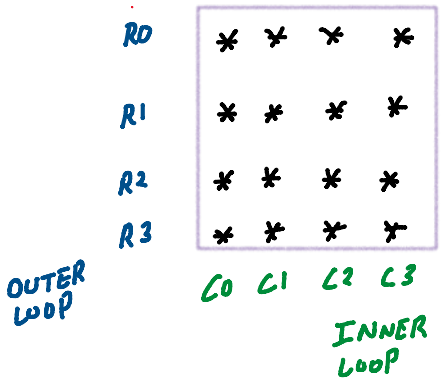
HW CLASS:03 __ 28/08/23

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PATTERN PROBLEMS

Step 01: Count the numbers of row (outer loop)
Step 02: See what is happening in each row (inner loop)

PATTERN:01 Square pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 4 (row < 4)

Step 02: See what is happening in each row (inner loop)

Row0 = print 4 star
Row1 = print 4 star
Row2 = print 4 star
Row3 = print 4 star

(col < 4)

CODE OF PATTERN 01

```cpp
#include<iostream>
using namespace std;

int main(){
    int n = 4;

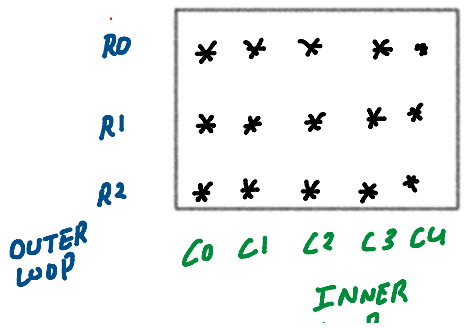
    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to n-1
        for(int col = 0; col < n; col = col + 1){
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

DRY RUN OF PATTERN 01

n = 4

row	row < n	col	col < n	print *	col = col+1	next line	row = row+1
0	0 < 4	0	0 < 4	*	1		
		1	1 < 4	*	2		
		2	2 < 4	*	3		
		3	3 < 4	*	4		
		4	4 < 4 → END	—	—	✓	1
1	1 < 4	0	0 < 4	*	1		
		1	1 < 4	*	2		
		2	2 < 4	*	3		
		3	3 < 4	*	4		
		4	4 < 4 → END	—	—	✓	2
2	2 < 4	0	0 < 4	*	1		
		1	1 < 4	*	2		
		2	2 < 4	*	3		
		3	3 < 4	*	4		
		4	4 < 4 → END	—	—	✓	3
3	3 < 4	0	0 < 4	*	1		
		1	1 < 4	*	2		
		2	2 < 4	*	3		
		3	3 < 4	*	4		
		4	4 < 4 → END	—	—	✓	4
4	4 < 4 → Outer loop ending						

PATTERN:02 Rectangle pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 3 (row < 3)

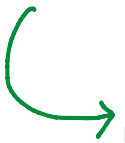
Step 02: See what is happening in each row (inner loop)

OUTER LOOP

C0 C1 C2 C3 C4

INNER LOOP

Step 02: See what is happening in each row (inner loop)



Row0 = print 5 star
Row1 = print 5 star
Row2 = print 5 star

(col < (3+2))

CODE OF PATTERN 02

```cpp
#include<iostream>
using namespace std;

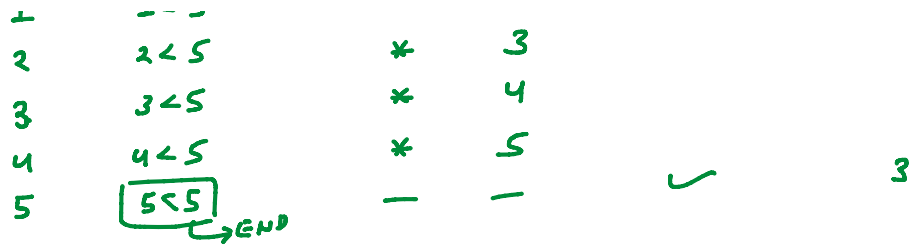
int main(){
    int n = 3;

    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to n+1
        for(int col = 0; col < (n+2); col = col + 1){
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

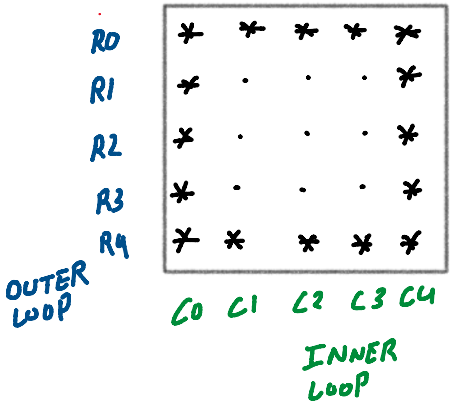
DRY RUN OF PATTERN 02

n = 3

row	row < n	col	col < (n+2)	Print *	col = col + 1	next line	row = row+1
0	0 < 3	0	0 < 5	*	1		
		1	1 < 5	*	2		
		2	2 < 5	*	3		
		3	3 < 5	*	4		
		4	4 < 5	*	5	✓	1
		5	5 < 5 → END	—	—		
1	1 < 3	0	0 < 5	*	1		
		1	1 < 5	*	2		
		2	2 < 5	*	3		
		3	3 < 5	*	4		
		4	4 < 5	*	5	✓	2
		5	5 < 5 → END	—	—		
2	2 < 3	0	0 < 5	*	1		
		1	1 < 5	*	2		
		2	2 < 5	*	3		
		3	3 < 5	*	4		



PATTERN:03 Hollow rectangle pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 5 (row < 5)

Step 02: See what is happening in each row (inner loop)

Row0 = print 5 star
Row1 = print 1* 3space 1*
Row2 = print 1* 3space 1*
Row3 = print 1* 3space 1*
Row4 = print 5 star

if (r0 || r4)
↳ print 5 time *
if (c0 || c4)
↳ print 1 time *
else
↳ print 3 time space

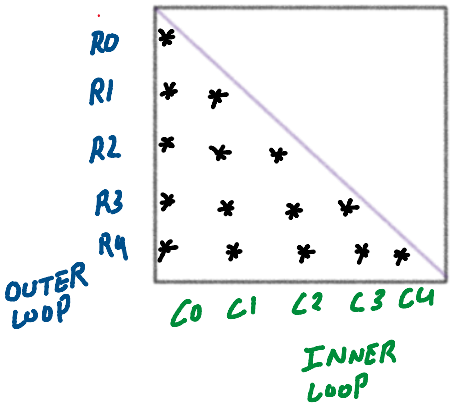
CODE OF PATTERN 03

```cpp
#include<iostream>
using namespace std;

int main(){
    int n = 5;

    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to n-1
        for(int col = 0; col < n; col = col + 1){
            // if zeroth or last row, then print n times star
            if(row == 0 || row == n-1){
                cout << "* ";
            }
            else{
                // if zeroth or last col, then print 1 time star
                if(col == 0 || col == n-1){
                    cout << "* ";
                }
                else{
                    cout << "  ";
                }
            }
        }
        cout << endl;
    }
    return 0;
}
```

PATTERN:04 Half Pyramid pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 5 (row < 5)

Step 02: See what is happening in each row (inner loop)

Row0 = print 1 star
Row1 = print 2 star
Row2 = print 3 star
Row3 = print 4 star
Row4 = print 5 star

(col < row+1)

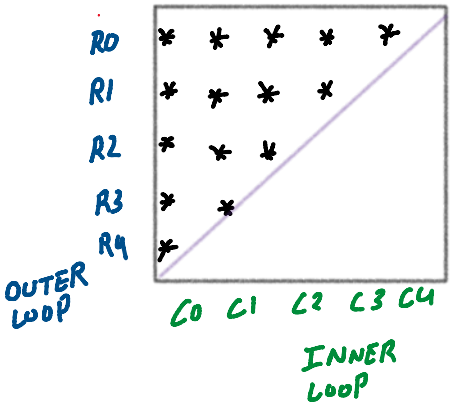
CODE OF PATTERN 04

```cpp
#include<iostream>
using namespace std;

int main(){
    int n = 5;

    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to (row+1)-1
        for(int col = 0;col < (row + 1); col = col + 1){
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

PATTERN:05 Inverted half pyramid pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 5 (row < 5)

Step 02: See what is happening in each row (inner loop)

Row0 = print 5 star
Row1 = print 4 star
Row2 = print 3 star
Row3 = print 2 star
Row4 = print 1 star

(col < (5-row))

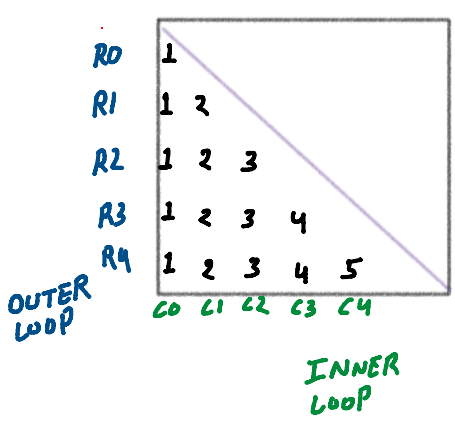
CODE OF PATTERN 05

```cpp
#include<iostream>
using namespace std;

int main(){
    int n = 5;

    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to n-row
        for(int col = 0; col < (n - row); col = col + 1){
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

PATTERN:06 Numerical half pyramid pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 5 (row < 5)

Step 02: See what is happening in each row (inner loop)

Row0 = print 1
Row1 = print 1-2
Row2 = print 1-3
Row3 = print 1-4
Row4 = print 1-5

(col < (row+1))

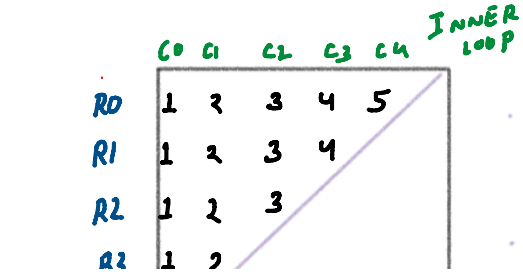
CODE OF PATTERN 06

```cpp
#include<iostream>
using namespace std;

int main(){
    int n = 5;

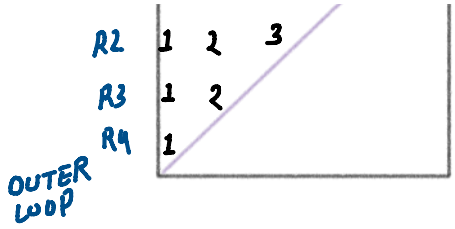
    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to (row+1)-1 and print col+1
        for(int col = 0; col < (row + 1); col = col + 1){
            cout << col+1 << " ";
        }
        cout << endl;
    }
    return 0;
}
```

PATTERN:07 Inverted numerical half pyramid pattern



Step 01: Count the numbers of row (outer loop)

Numbers of row = 5 (row < 5)



Numbers of row = 5 (row < 5)

Step 02: See what is happening in each row (inner loop)

Row0 = print 1-5
Row1 = print 1-4
Row2 = print 1-3
Row3 = print 1-2
Row4 = print 1

(col < (5-row))

CODE OF PATTERN 07

```cpp
#include<iostream>
using namespace std;

int main(){
    int n = 5;

    // outer loop traverse from 0 to n-1
    for(int row = 0; row < n; row = row + 1){
        // inner loop traverse from 0 to n-row and print col+1
        for(int col = 0; col < (n-row); col = col +1){
            cout << col+1 << " ";
        }
        cout << endl;
    }
    return 0;
}
```

All flowcharts are converted into C++ programs

```cpp
// Program01: Multiply two numbers by taking input from user
#include<iostream>
using namespace std;

int main(){
    int a, b;

    cout << "Enter a" << endl;
    cin >> a;

    cout << "Enter b" << endl;
    cin >> b;

    int result = a * b;
    cout << "Multiply of a and b is " << result << endl;

    return 0;
}

/*
OUTPUT:
Enter a
2
Enter b
3
Multiply of a and b is 6
*/
```

Linkedin: @manojofficialmj

```cpp
// Program02: Find the perimeter of a triangle
#include<iostream>
using namespace std;

int main(){
    int a = 3, b = 4, c = 5;
    int p = a + b + c;
    cout << "Piremeter of a triangle is " << p << endl;
    return 0;
}

/*
OUTPUT:
Piremeter of a triangle is 12
*/
```

Linkedin: @manojofficialmj

```cpp
// Program03: Find the simple interest
#include<iostream>
using namespace std;

int main(){
    int p = 100, r = 2, t = 1;
    double SI = (p*r*t)/100;
    cout << "Simple interest is " << SI << endl;
    return 0;
}

/*
OUTPUT:
Simple interest is 2
*/
```

Linkedin: @manojofficialmj

```cpp
// Program04: Find the compound interest
#include <bits/stdc++.h>
#include<iostream>
using namespace std;

int main()
{
    double p = 10000, r = 5, t = 2;
    double A = p * (pow((1 + r / 100), t));
    double CI = A - p;
    cout << "Compound interest is " << CI;
    return 0;
}

/*
OUTPUT:
Compound interest is 1025
*/
```

Linkedin: @manojofficialmj

```cpp
// Program05: Print counting from n to 1
#include<iostream>
using namespace std;

int main(){
    int n, counting = 0;
    cout << "Enter a number" << endl;
    cin >> n;
    for(int i = n; i > 0; i = i - 1){
        counting = counting + i;
    }
    cout << "Counting is " << counting << endl;
    return 0;
}

/*
OUTPUT:
Enter a number
3
Counting is 6
*/
```

Linkedin: @manojofficialmj

```cpp
// Program06: Find the factorial of a number
#include<iostream>
using namespace std;

int main(){
    int n, fac = 1;
    cout << "Enter a number " << endl;
    cin >> n;
    if(n < 0){
        cout << "Invalid input" << endl;
    }
    else if(n == 0){
        cout << "Factorial of " << n << "! is " << fac << endl;
    }
    else if(n > 0){
        for( n ; n > 0; n = n - 1 ){
            fac = fac * n;
        }
        cout << "Factorial of " << n << "! is " << fac << endl;
    }
    return 0;
}

/*
OUTPUT:
Enter a number
-1
Invalid input

Enter a number
0
Factorial of 0! is 1

Enter a number
5
Factorial of 5! is 120
*/
```

Linkedin: @manojofficialmj

```cpp
// Program07: Check if number is prime or not
#include<iostream>
using namespace std;

int main(){
    int n;
    cout << "Enter a number" << endl;
    cin >> n;

    if(n<2){
        cout << n << " is not prime number" << endl;
    }
    else{
        for(int i = 2; i < n; i = i + 1){
            if(n % i == 0){
                cout << n << " is not prime number" << endl;
            }
        }
        cout << n << " is prime number" << endl;
    }
    return 0;
}

/*
OUTPUT:
Enter a number
-1
-1 is not prime number

Enter a number
0
0 is not prime number

Enter a number
1
1 is not prime number

Enter a number
2
2 is prime number
*/
```

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```cpp
// Program08: Check valid triangle or not
#include<iostream>
using namespace std;

int main(){
    int a,b,c;

    cout << "Enter side a = ";
    cin >> a;
    cout << "Enter side b = ";
    cin >> b;
    cout << "Enter side c = ";
    cin >> c;

    if(a+b > c && a+c > b && b+c > a){
        cout << "Valid triangle" << endl;
    }
    else{
        cout << "Invalid triangle" << endl;
    }
    return 0;
}

/*
OUTPUT:
Enter side a = 3
Enter side b = 4
Enter side c = 5
Valid triangle

Enter side a = 0
Enter side b = 2
Enter side c = 1
Invalid triangle
*/
```

Linkedin: @manojofficialmj

```cpp
// Program09: Print max of three numbers
#include<iostream>
using namespace std;

int main(){
    int a,b,c;
    cout << "Enter number a = ";
    cin >> a;
    cout << "Enter number b = ";
    cin >> b;
    cout << "Enter number c = ";
    cin >> c;

    if(a>b && a>c){
        cout << "Maximum number is a" << endl;
    }
    else if(b>a && b>c){
        cout << "Maximum number is b" << endl;
    }
    else{
        cout << "Maximum number is c" << endl;
    }
}

/*
OUTPUT:
Enter number a = 1
Enter number b = 2
Enter number c = 3
Maximum number is c
*/
```

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